LEMUEL PRANOTO

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Software Engineer

/lemuelpranoto
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SKILLS

Programming Languages Java, Scala, JavaScript, TypeScript, HTML, CSS, PHP, C#, C, C++, Python, R, Verilog

Frameworks & Technologies Amazon Web Services (AWS), Docker, Kubernetes, Jenkins, Kafka, Nx, Angular, NestJS, React, React Native, Node.js, Express.js, REST APIs, TypeORM, Jest, Mockito, PostgreSQL, MongoDB, Redis, Postman, Git, GitHub Actions, IntelliJ, VS Code, Visual Studio, Jira, Figma, OpenGL, .NET

WORK EXPERIENCE

Scription **April 2025 – October 2025**
Intermediate Software Engineer *Calgary, AB*

- Engineered and maintained a full-stack web platform using **Angular**, **NestJS**, and an **Nx monorepo**, improving modularity and enabling reusable shared libraries across multiple services and applications.
- Optimized **GitHub Actions CI/CD pipelines** by implementing caching, Nx affected builds, parallelized test and lint jobs, and custom tooling to reduce build and test execution times in a multi-project repository.
- Implemented custom **ESLint rules** and code quality tooling, managed and executed **TypeORM database migrations**, and expanded **Jest unit and integration test coverage** to ensure consistent code quality and reliable production deployments.

Surya Titia Makmur **April 2023 – May 2024**
Full-Stack Developer

- Designed and developed a full-stack product and inventory management web app using **React** and **Node.js** with **Express.js**.
- Integrated backend services with RESTful APIs, authentication, routing, and middleware to support secure backend operations.
- Spearheaded migration from spreadsheet-based storage to **DynamoDB** by designing a scalable data model and implementing Python-based **data ingestion** pipelines with batch processing, improving query performance by 40%.

Amazon.com, Inc. **August 2022 – April 2023**
Software Development Engineer *Vancouver, BC*

- Designed and developed a new high-traffic payment authorization authentication widget at retail checkout by writing a dynamic **idempotency key** retrieval logic code in **Scala** alongside fetching other important payment contract metadata from external PayStation entity to successfully onboard a new retail EBT payment method.
- Established a **Java GET SPI** as part of a legacy decoupled payment method **(DPM)** migration that requests payment contract information from external Acculynk PayStation service then returns corresponding data reliably in a high-traffic environment.
- Leveraged native **AWS** technologies to streamline modular **CI/CD** workflows toward sprints and milestones while employing **Amazon Ion** and **RDE stack** to conduct rigorous unit/integration testing, ensuring that code packages and environments shipped to production are robust, maintainable and secure.
- Handled high-severity, customer-impacting tickets during **oncall** by performing **root-cause analysis** of logs and metrics, implementing new metrics and alarms to monitor services and improve **debugging productivity** of fellow service-dependent up/downstream developers and process engineers alike.

SeaHawk Robotics **June 2021 – August 2021**
Software Engineer Intern

- Built an FPV drone GUI in **Mission Planner GCS** and optimized live video stream from four onboard camera sources by refining latency and resolution with **GStreamer** resulting in an improved stream quality resolution and a 40% reduction in pipeline latency.
- Constructed a winch control interface by modifying the existing control firmware script in **Python** to factor in depth feedback relayed from drone sensors when calculating winch rotation and depth of payload drop.

Pinnacle Investments **May 2020 – August 2020**
Software Engineer Intern

- Deployed an **AWS Lambda Function** GET request **REST API** endpoint that queries a **PostgreSQL** database by utilizing **SQL** statements to retrieve data based on the parameters set on the Portfolio Management Summary (PMS) mobile app.
- Streamlined a reduction in PMS database query time and established an in-house **stock market API** server for their new branch.

EDUCATION

University of British Columbia ***Vancouver, BC***
Bachelor of Applied Science, Computer Engineering with Co-op *2018-2022*
Awards: Capstone UBC ECE Faculty Award Winner, 2x Outstanding Student Scholarship, 4x Honorable Mention (Term GPA > 3.7)

TECHNICAL PROJECTS

Land Monitor Mobile-App **September 2021 – April 2022**
KoroTU Technology (*Capstone Project*)

- Designed and developed a **hybrid mobile-app** that detects and informs users of deforestation changes using Land Use/Land Cover (LULC) classification algorithms, utilizing **React Native** to support cross-platform **iOS** and **Android** devices during front-end architectural app build, **Figma** to prototype the user interface design and **Expo** to build and deploy latest version iteratively.
- Architected map component utilizing **Google Earth Engine API** alongside **Google Cloud Storage** for image retrieval and processing, **AWS** for back-end server hosting the database, user authentication data, and search results caching; successfully deployed through the rigorous testing of **REST API endpoints** with **Postman** and **Docker** to ensure environment compatibility.